Juan Tomás Martínez

Forward Deployed Engineer · Requirements, solution architecture and AI-assisted delivery

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PROFILE

I take unclear client problems all the way to solutions in production. I gather requirements, design the solution and the data model, and build it by directing AI coding agents (Claude Code) through specs, TDD and review of every change. That's how I took AudenPass to production with Desarrollos Guía, and how I work with enterprise clients at Aeroterra.

EXPERIENCE

Co-founder · Solutions Architect & Backend Engineer — Desarrollos Guía Aug 2026 – Present

desarrollosguia.com — custom software studio. Main client: AudenPass (audenpass.com), a ticketing platform live in production with real payments.

- Gathered requirements with the client, wrote the spec with acceptance criteria, and designed the backend architecture and data model (16 tables).
- Built the backend (Node.js, TypeScript, Express, PostgreSQL, Redis) by directing AI coding agents (Claude Code with subagents) through TDD and review of every change: 50+ REST endpoints in OpenAPI and ~750 automated tests.
- Integrated Mercado Pago payments (Orders API) with HMAC-verified webhooks, idempotency and periodic reconciliation from a background worker.
- Designed stock control with PostgreSQL transactions and locks to prevent overselling across replicas; validated it with a local k6 load test.
- Deployed on AWS (Lightsail, SES, least-privilege IAM) with GitHub Actions: database snapshot before migrating, zero-downtime rollout and rollback.

Functional Analyst / Solutions Engineer — Aeroterra Apr 2025 – Present

Buenos Aires. Enterprise projects for clients in energy, government and public safety.

- Clients under NDA; scope, volumes and names are not disclosed.
- First point of contact between clients and development teams: I gather, analyze and document requirements and turn them into workable solutions.
- Designed a 3-level relational data model (report → event → follow-up) and a standalone inspection module for an oil & gas client's field-operations platform.
- Use Python daily (ArcGIS API for Python, Jupyter notebooks and scripts) to automate the creation and management of data services.
- Present solutions through demos and proofs of concept, and validate integrations by testing REST APIs with Postman.

PROJECTS

- **InvGate → Jira** — turns ITSM ticket PDFs into Jira-ready user stories with acceptance criteria. Built with Claude Code.
- **Personal finance assistant** — LLM API integrated into my own code: prompts, token handling and structured outputs.

SKILLS

Analysis & design: requirements gathering, user stories and acceptance criteria, UML, relational data modeling, solution architecture, demos and proofs of concept

Development: Python (daily), TypeScript, JavaScript, SQL · Node.js, Express, FastAPI · React, Next.js · REST APIs, OpenAPI · PostgreSQL, Redis

AI: building by directing AI coding agents (Claude Code with subagents, TDD) · LLM API integration (prompts, structured outputs)

Cloud & CI/CD: AWS (Lightsail, SES, IAM) · GitHub Actions · Docker / Docker Compose · Cloudflare

Integrations, testing and platforms: Mercado Pago (webhooks, idempotency) · Postman · Vitest, Testcontainers, k6 · ArcGIS Enterprise (Survey123, Field Maps, Dashboards)

EDUCATION & LANGUAGES

University Technician in Systems Programming · 2023 – 2024 · Spanish (native) · Professional English (CILE 3, UBA)